

Question 1 – Online Shopping System

Create the following tables:

Customer

customerid, customername, email, phone

Orders

orderid, orderdate, amount, status, customerid

A. CREATE TABLE

1. Create Customer with customerid as the **PRIMARY KEY**.
2. customername should be **NOT NULL**.
3. email should be **NOT NULL** and **UNIQUE**.
4. phone should be **UNIQUE**.
5. Create Orders with orderid as the **PRIMARY KEY**.
6. orderdate should be **NOT NULL**.
7. Set the default value of amount to 0.
8. Set the default value of status to 'Pending'.
9. customerid should be a **FOREIGN KEY** referencing Customer(customerid).

B. ALTER TABLE

10. Add a new column city to the Customer table.
11. Add a new column shippingaddress to the Orders table.
12. Add a **NOT NULL** constraint to shippingaddress.
13. Add a new column paymentmode to the Orders table.
14. Set the default value of paymentmode to 'Cash'.
15. Rename the phone column of Customer to mobileno if your database supports column renaming.

C. INSERT

16. Insert **5 customers** into the Customer table.
17. Insert **7 orders** into the Orders table.
18. Insert one order without specifying status and verify that the default value is inserted.
19. Insert one order without specifying amount and verify the default value.

D. UPDATE

20. Update the city of a particular customer.
21. Increase the amount of a particular order.
22. Change the status of an order from 'Pending' to 'Confirmed'.
23. Change the payment mode of a particular order.

E. DELETE

24. Delete a particular order.
25. Delete a customer who does not have any orders.
26. Try deleting a customer who has orders and observe the result because of the **FOREIGN KEY** constraint.

Question 2 – Movie Ticket Booking System

Create the following tables:

Movie

`movieid, moviename, language, duration`

Booking

`bookingid, customername, ticketcount, bookingdate, movieid`

A. CREATE TABLE

1. Create `Movie` with `movieid` as the **PRIMARY KEY**.
2. `moviename` should be **NOT NULL** and **UNIQUE**.
3. `language` should be **NOT NULL**.
4. Create `Booking` with `bookingid` as the **PRIMARY KEY**.
5. `customername` should be **NOT NULL**.
6. Set the default value of `ticketcount` to 1.
7. `bookingdate` should be **NOT NULL**.
8. `movieid` should be a **FOREIGN KEY** referencing `Movie(movieid)`.

B. ALTER TABLE

9. Add `price` to the `Booking` table.
10. Set the default value of `price` to 200.
11. Add `theatre` to the `Booking` table.
12. Add a **NOT NULL** constraint to `theatre`.
13. Add `genre` to the `Movie` table.
14. Add a **UNIQUE** constraint on `theatre` **only if appropriate for your design**. Explain why it may or may not be suitable.

C. INSERT

15. Insert **5** movies.
16. Insert **8** bookings.
17. Insert a booking without specifying `ticketcount` and verify the default value.
18. Insert a booking without specifying `price` and verify the default value.

D. UPDATE

19. Change the language of a movie.
20. Update the duration of a movie.
21. Change the number of tickets for a particular booking.
22. Change the price of a booking.
23. Change the theatre of a booking.

E. DELETE

24. Delete a particular booking.
 25. Delete a movie that has no bookings.
 26. Try deleting a movie that is referenced by a booking and observe the result.
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Question 3 – Vehicle Service Center

Create the following tables:

Vehicle

vehicleid, vehiclenu, model, ownername, phone

Service

serviceid, servicedate, servicetype, charges, vehicleid

A. CREATE TABLE

1. Create Vehicle with vehicleid as the **PRIMARY KEY**.
2. vehiclenu should be **NOT NULL** and **UNIQUE**.
3. ownername should be **NOT NULL**.
4. phone should be **UNIQUE**.
5. Create Service with serviceid as the **PRIMARY KEY**.
6. servicedate and servicetype should be **NOT NULL**.
7. Set the default value of charges to 0.
8. Create a **FOREIGN KEY** on vehicleid referencing Vehicle(vehicleid).

B. ALTER TABLE

9. Add color to the Vehicle table.
10. Add address to the Vehicle table.
11. Add servicestatus to the Service table.
12. Set the default value of servicestatus to 'Pending'.
13. Add mechanicname to the Service table.
14. Modify the size of ownername to allow longer names.
15. Add a **UNIQUE** constraint on phone if it was not defined while creating the table.

C. INSERT

16. Insert **5 vehicles**.
17. Insert **8 service records**.
18. Insert a service record without specifying `charges` and verify the default value.
19. Insert a service record without specifying `servicestatus` and verify the default value.

D. UPDATE

20. Change the color of a particular vehicle.
21. Change the owner's phone number.
22. Increase the service charges for a particular service.
23. Change `servicestatus` from 'Pending' to 'Completed'.
24. Change the mechanic assigned to a service.

E. DELETE

25. Delete a particular service record.
26. Delete a vehicle that has no service records.
27. Try deleting a vehicle that has service records and observe the FOREIGN KEY constraint.